Example 3.15

A cricket team won two matches more than they lost. If they win they get 5 points and for loss -3 points, how many matches have they played if their total score is 50.



*Fig.*3.10

Solution

Let the number of matches lost $= x$.

Then number of matches won $= x + 2$.

Given that, $5(x + 2) + (-3)x = 50$

$5x + 10 - 3x = 50$

$2x + 10 = 50$

$2x + 10 - 10 = 50 - 10$ (Subtract 10 on both sides)

$2x = 40$

Divide by 2 on both the sides, $x = 20$.

Therefore, the number of matches played is $x + x + 2 = 2x + 2$

$= (2 \times 20) + 2$

$= 40 + 2$

$= 42$.

Try this

Kandhan and Kavya are friends. Both of them are having some pens.



Kandhan : If you give me one pen, then we will have equal number of pens. Will you?

Kavya : But, if you give me one of your pens, then mine will become twice as yours. Will you?

Construct algebraic equations for this situation. Can you guess and find the actual number of pens, they have?

Exercise 3.3

1. Fill in the blanks.

   (i) An expression equated to another expression is called _____

   (ii) If $a = 5$, the value of $2a + 5$ is _____

   (iii) The sum of twice and four times of the variable x is _____

2. Say True or False.

   (i) Every algebraic expression is an equation

   (ii) The expression $7x + 1$ can not be reduced without knowing the value of x.

   (iii) To add two like terms, its coefficients can be added.

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